

Mechanistic Modeling of Preparative Column Chromatography for Biotherapeutics

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Abstract

Chromatography has long been, and remains, the workhorse of downstream processing in the production of biopharmaceuticals. As bioprocessing has matured, there has been a growing trend toward seeking a detailed fundamental understanding of the relevant unit operations, which for some operations include the use of mechanistic modeling in a way similar to its use in the conventional chemical process industries. Mechanistic models of chromatography have been developed for almost a century, but although the essential features are generally understood, the specialization of such models to biopharmaceutical processing includes several areas that require further elucidation. This review outlines the overall approaches used in such modeling and emphasizes current needs, specifically in the context of typical uses of such models; these include selection and improvement of isotherm models and methods to estimate isotherm and transport parameters independently. Further insights are likely to be aided by molecular-level modeling, as well as by the copious amounts of empirical data available for existing processes.

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INTRODUCTION

Column chromatography is the undisputed leader in achieving high-selectivity purification of proteins and has consequently become the workhorse for purification of protein therapeutics over the last two decades (1, 2). Chromatographic unit operations are used for capture, intermediate polishing, and final polishing steps (3), which in most cases are executed in batch mode (4). Continuous processes—typically operated as cyclic multicolumn operations—are more challenging to operate and are still the subject of development efforts, but they are gradually becoming more common in the biopharmaceutical industry (5–7). In either batch or continuous operation, chromatography appears likely to retain its dominant role in biopharmaceutical purification despite the ongoing quest for nonchromatographic alternatives (8–10).

Although the batch chromatographic processes that still dominate biopharmaceutical production have become very efficient, chromatographic process development is still not highly streamlined. Many biotherapeutic protein purification schemes, especially for monoclonal antibodies (mAbs), follow a platform process approach (11), but even with the less extensive parametric exploration required in such cases, the cost of development remains high. Although heuristic approaches can help in streamlining process development, quantitative modeling of chromatography processes can be more powerful in optimizing the operating conditions, and hence in helping to reduce operating costs (12). Process models can also help in improving product purity and throughput. The process know-how that models can provide can create a design space for process variables that ensures adequate product quality, in accordance with the quality-by-design principle required by regulatory agencies such as the US Food and Drug Administration (13).

Two modeling frameworks are currently widely used in chromatography process development. The first directly uses analyses of empirical data, based mainly on statistical methods such as the design of experiments (14, 15) or simplex design principles (16, 17) that are commonly used in process characterization and optimization. These models do not require a detailed understanding of the process but can still capture the critical quality attributes of the product within the design space of the model. Such approaches are suitable for establishing causal relationships in complex processes for which mechanistic models are unavailable, or indeed where mechanistic understanding is not yet adequate to allow such models to be developed.

The second approach uses deterministic models that incorporate mechanistic descriptions of solute transport (inter- and intraparticle) and adsorption (18, 19). Efforts to model chromatographic or other adsorptive processes date back approximately a century (20), so it is reasonable to expect that the mechanistic understanding of chromatography, both qualitative and quantitative, has achieved a sufficient level of maturity to favor such models over the empirical ones mentioned above. The purpose of this review is to assess the current state of knowledge of such mechanistic models specifically for applications in biopharmaceutical purification. To this end, we address the general principles incorporated in such models, as well as recent applications to various relevant modes of chromatography, including affinity, ion-exchange, hydrophobic interaction, and multimodal chromatography. The transport and adsorption phenomena involved are qualitatively largely independent of the mode of chromatography, but the contributions of different mechanisms may vary both with the mode of chromatography and with the adsorbent architecture (21). Although the molecular-level basis for adsorption varies among different chromatographic modes, these differences are often obscured by the use of generic isotherm models. However, the differences among the modes add to the challenges of model calibration and parameter determination, which remain a bottleneck for the adoption of models for process optimization and development. We therefore also briefly discuss adsorption models based on molecular-level descriptions. The application of multicomponent models for separation of impurities is discussed, as is model-based process control.

The transport and adsorption elements of a chromatography model identified above must be combined into what is necessarily a multiscale model—column and particle scales—even in the absence of molecular-level detail. Furthermore, the high solute loadings needed for high-productivity preparative purification introduce nonlinearities via the adsorption isotherm. These features have been incorporated into models in different degrees of rigor over the past century, although the features or approximations included have often been selected based on their impact on the tractability of the model equations in analytical form. Despite these constraints, some of the most important aspects of modern model predictions emerged from early work; for example, DeVault's (22) column model assuming local equilibrium, without any intraparticle details, allows prediction of such phenomena as self-sharpening fronts that give rise to constant-pattern solutions.

The multiscale nature of the models is readily apparent in extensions of solely column-scale models to include the particle scale, which may be done purely functionally or also structurally. The earliest functional models were kinetic (20, 23), but the engagement of chemical engineers soon led to accounting for mass-transfer limitations. These models differed in the use of lumped mass-transfer descriptions, with either liquid (24) or solid (25) linear driving forces, or more rigorous diffusion models postulating pore (26) and/or homogeneous ("solid"/surface) diffusion (26, 27). The Langmuir isotherm was a frequent choice (26), although mathematical tractability often required the limiting rectangular-isotherm approximation (28).

The current consensus is that an appropriate chromatographic model consists, at the column level, of a hyperbolic partial differential equation describing axial transport in the mobile phase, and at the particle level, of additional differential equation(s) governing intraparticle diffusion and adsorption kinetics and equilibrium (29). Specification of the conditions at the column inlet allows, in principle, solution for the conditions at the column outlet. Analytical solutions, especially for nonlinear isotherms, are generally not available, and the approximate solutions developed in the mid-twentieth century are unsuitable for quantitative predictions, so numerical solutions are necessary. The surge in interest in detailed modeling of biopharmaceutical separations has catalyzed the development of multiple software packages that allow simulations to be performed efficiently and accurately. Although these packages resolve the mathematical challenge of solving the governing equations, they are best viewed as enabling tools in the quest for effective models; the other key ingredient is a thorough understanding of protein chromatographic science. We seek to address both of these aspects below.

TRANSPORT MODELS FOR PACKED-BED CHROMATOGRAPHY

The most comprehensive model that includes all of the commonly accepted phenomena for the transport behavior of a protein within a bed of porous spherical particles packed in a cylindrical column is the general rate model (GRM) (30). The model accounts for the transport of protein in the interstitial space by convection and axial dispersion; mass transfer of solute from the interstitial space to the particle surface and onward into the static fluid in the intraparticle pore space; diffusion within the particle by pore and/or homogeneous diffusion; and finally, transfer of protein to the adsorbed state, normally governed by a kinetic term (**Figure 1**). The transport phenomena are captured by the equations for the GRM (30):

$$\frac{\partial c_i}{\partial t} = -u \frac{\partial c_i}{\partial z} + D_{ax,i} \frac{\partial^2 c_i}{\partial z^2} - \frac{1 - \epsilon_c}{\epsilon_c} \frac{3}{r_p} k_{film,i} [c_i - c_{pi}(\cdot, \cdot, r_p)] \text{ for } t \geq 0, z \in [0, L] \quad 1.$$

$$\begin{aligned} \frac{\partial c_{pi}}{\partial t} = & D_{pi} \left(\frac{\partial^2 c_{pi}}{\partial r^2} + \frac{2}{r} \frac{\partial c_{pi}}{\partial r} \right) + D_{si} \frac{1 - \epsilon_p}{\epsilon_p} \left(\frac{\partial^2 q_i}{\partial r^2} + \frac{2}{r} \frac{\partial q_i}{\partial r} \right) \\ & - \frac{1 - \epsilon_p}{\epsilon_p} \frac{\partial q_i}{\partial t} \text{ for } t \geq 0, z \in [0, L] \text{ and } r \in [0, r_p]. \end{aligned} \quad 2.$$

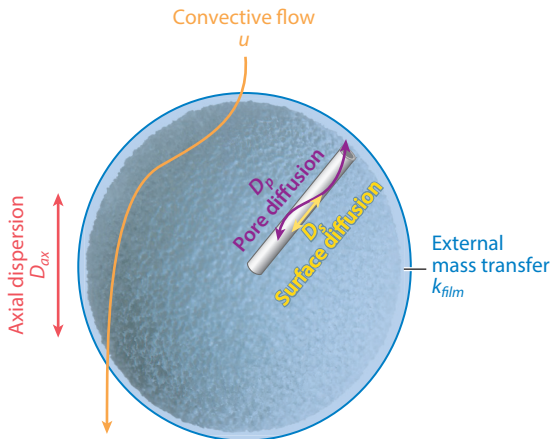


Figure 1

Schematic of transport processes outside and inside a porous resin bead.

The first equation describes the concentration c_i of component i in the mobile phase (interstitial space) with a linear velocity u , subject to axial dispersion (coefficient D_{ax}) and mass transfer to the particle surface (mass transfer coefficient $k_{film,i}$) for spherical beads of radius r_p . The second equation describes the concentration c_{pi} of the solute within the pore space and its adsorption on the solid (concentration q_i). Both the free and adsorbed protein can diffuse, with pore and homogeneous diffusivities D_p and D_s , respectively. The column (interstitial) void fraction is ϵ_c , and the intraparticle (pore) void fraction is ϵ_p . The model is coupled with a kinetic model to describe adsorption, the details of which vary with the different modes of chromatography or the isotherm model. The Danckwerts boundary conditions at the column inlet and outlet are most commonly used with Equation 1, and the boundary conditions at the surface of the bead, used with Equation 2, account for the mass balances at the outer bead surface,

$$k_{film,i}[c_i - c_{p,i}(\cdot, \cdot, r_p)] = \epsilon_p D_{p,i} \frac{\partial c_{p,i}}{\partial r}(\cdot, \cdot, r_p) \text{ for } t \geq 0 \text{ and } z \in [0, L], \quad 3.$$

and symmetry at the bead center

$$\frac{\partial c_{p,i}}{\partial r}(\cdot, \cdot, 0) = 0 \text{ for } t \geq 0 \text{ and } z \in [0, L]. \quad 4.$$

Application of these equations of course requires specification of the values of the different parameters, and more subtly also accounting for the possible dependence of those parameter values on the local solution conditions. Most importantly, the adsorption phenomena, which ultimately drive chromatographic separation, must be accommodated in a separate equation, which can be expressed in the form of an equilibrium isotherm or the underlying kinetic terms. This element of the process incorporates additional parameters that are especially important to the overall description. Taken together, the parameters characterizing the various phenomena result in many degrees of freedom, which can lead to confounding effects if parameter estimation is performed by fitting of solutions of the full model to column chromatographic data. However, the ability to specify parameter values from independent measurements greatly enhances the value of the model and is preferred in enabling truly predictive calculations.

In addition to the GRM, other, simpler representations of chromatographic phenomena are also used regularly. Each of these incorporates one or more forms of lumping of relevant

phenomena, making predictive use less practicable. These models are discussed in standard textbooks (29–31), and only a few examples are briefly considered here. The simplest of these models for which reasonable column simulations can be achieved is the equilibrium dispersive model (EDM) (32), but it lacks intraparticle transport limitations and so is inadequate for detailed calculations. Other lumped models exceed the EDM in rigor and are more widely used, e.g., the transport-dispersive model (TDM) or the reaction-dispersive model. These models are employed to reduce the effort in solving the model equations while retaining some representation of mass-transfer limitations in the resin bead. The TDM is essentially the GRM, simplified by lumping all the mass transfer resistance into a linear driving-force approximation, with a uniform concentration implicitly assumed inside the particle (33–35). In recent work, the TDM has also been modified further in assuming the mass-transfer resistance in the solid phase to be dominant in calculating predicted elution peak profiles (36–38). The reaction dispersive model lumps adsorption and desorption kinetics with a linear mass-transfer driving force description, leading to a form that is mathematically equivalent to the TDM (39). Although such models continue to be used to predict elution peak profiles, the lumping can lead to erroneous descriptions of intraparticle concentrations, leading to significant errors, e.g., for binary systems with intraparticle displacement (40).

One area in which simplified descriptions may be valuable is in estimating at least a subset of the model parameters based on independent experiments that are consistent with the simplified models. An example is that peak retention can be predicted in the limit of a linear isotherm using analytical solutions or moment analysis, allowing parameters to be estimated for the linear region of the isotherm. Miyabe (41) provided a detailed overview of moment analysis to analyze peak shapes from a kinetic point of view, where the moments are derived by Laplace or Hankel transforms of the GRM. Although the first moment of the peak, representing the retention time, can usually be determined quite accurately, the transport properties can be ascertained only from the second or higher moments, which are often erroneous owing to detector noise. Nonetheless, these methods can be used to verify numerical simulations of the GRM (42), including determining the appropriate transport model and parameter values for conditions in the linear region of the isotherm under isocratic conditions (43). An extension especially useful for biotherapeutics, for which separations are typically performed in bind-and-elute mode with gradients in elution modifiers, is Yamamoto's so-called GH-IR curve, based on moment analysis with a linear isotherm. This employs a plot of the normalized gradient slope against the peak salt concentration to estimate the peak retention in ion-exchange chromatography (IEX) (44).

All of the above equations are for the 1D axial dimension of the column only; a more general scenario would be to also include radial mass transfer in the column, which is more important for large-scale columns, where the diameter of the column may be much larger than its length. A 2D form of the GRM has been formulated by considering different regions of injection at the inlet for core-shell particles (45), along with a numerical solution, including nonlinear isotherms for multicomponent mixtures. Several other aspects can also be incorporated in more complex forms; examples include constitutive relations for diffusion, e.g., the Maxwell–Stefan formulation (46), and the effects of a polydisperse size distribution in the resin particles (47). Although such complexities may indeed reflect physically meaningful situations, they present additional challenges in the need to estimate realistic values of any extra model parameters in the augmented models.

NUMERICAL SOLUTION OF THE GENERAL RATE MODEL

The unavailability of analytical solutions to the governing equations for the conditions that usually apply in protein separations has led to the widespread use of numerical solutions. Various numerical schemes and solution methods are used, depending on the accuracy or speed of the solution

required. The popular methods for spatial discretization of Equations 1 and 2 along the length of the column and the radius of the particles, respectively, are finite elements, finite differences, and finite volume schemes (29). The resulting ordinary differential equations are nonlinear and stiff. These implicit differential-algebraic equations can be solved using sparse matrix solvers.

Over the last decade, several software packages, both open source and commercial, have become available to carry out these calculations. The open-source CADET package employs a finite-volume scheme for Equations 1 and 2, with the concentrations at the cell boundaries approximated by a weighted, essentially non-oscillatory scheme to avoid traveling shock waves (48). The ChromX package employs the finite-element method, which is equally efficient in solving the GRM (49). Another method uses first-order accurate, finite-volume discretization (39) and a high-order accurate nodal discontinuous Galerkin finite-element method for solving the simpler EDM (50). A simple method using higher-order finite-difference together with an ordinary differential equation solver is also available (18), but these models do not explicitly decouple some of the mass-transfer kinetics and are useful only in determining the peak retention time.

ISOTHERM MODELS

The largest source of variability in chromatographic models, and correspondingly the one that lies at the heart of the adsorptive process, is the description of adsorption behavior as represented by the model used for adsorption and desorption kinetics or the equilibrium isotherm. This description of the distribution of protein between the liquid phase and the solid interface is especially challenging at high concentrations, where the full isotherm, including the plateau region, must be captured. Protein isotherms on chromatography media are generally hyperbolic, without the kinds of features related to multilayer adsorption that are incorporated in various classes of the Brunauer–Emmett–Teller classification.

Key features of protein adsorption isotherms are the initial slope, which reflects protein-surface affinity and hence, for competitive adsorption, determines adsorption selectivity, and the plateau region, which reflects adsorption capacity. The initial slope, despite characterizing the linear region that determines peak retention, is very difficult to estimate when the binding is very strong, as is often the case for proteins. Both the affinity and the capacity may be modulated by adjusting the solution conditions, using different modifiers in different modes, and this is the handle used to control elution and hence separation. As a result, the dependence of the isotherm parameters on solution conditions is an essential part of a predictive chromatographic model, beyond simply the dependence of the adsorbed concentration, q_i , on the solute concentration in the pore, c_{pi} . The isotherm models that are used often have some mechanistic basis, but the complexity of biotherapeutic proteins (anisotropic shape and charge distribution) and adsorbents makes such models based on structural information at best conceptually correct. The subsections below illustrate the issues for isotherm models for different modes of chromatography.

Isotherm Models for Ion-Exchange Chromatography

Ion exchange is the most widely used mode of chromatography for the production of biotherapeutics, and it has received the most attention in efforts to develop mechanistic models. In mAb production, cation exchange is used as a polishing step to remove product-related impurities such as protein aggregates and charge variants (51, 52), whereas anion exchange is often used in flow-through mode for host-cell protein and DNA removal (53). Accurate modeling of even single-component behavior in ion exchange is not yet consistently achieved, and multicomponent models that can also describe the behavior of a wide range of impurities, especially those not product related, are further afield yet because of the wide diversity but low concentrations of these species.

Among the many different models for ion exchange of proteins, the Langmuir and steric mass action (SMA) isotherms are the most popular. The mechanistic basis for the Langmuir model, specifically the assumption of well-defined, independent binding sites, is inconsistent with the physical scenario for proteins in IEX (54, 55). Although the isotherm shape is correctly represented qualitatively, analysis of batch isotherm data suggests that the Langmuir isotherm is softer—i.e., it has a less sharp transition to the plateau region—than is suitable for ion exchange and consequently often fails to fit experimental data adequately (56). This may be due to protein–protein repulsion, which cannot be ignored at higher loads.

The Langmuir model does not explicitly account for the dependence of adsorption on the concentration of salt, the usual mobile-phase modulator in IEX, so modifications have been introduced to the model parameters to reflect salt effects. These adjustments are variously based on model forms attributed to hydrophobic (57) or charge stoichiometric (58) effects, but they are not effective at all conditions, especially higher loads.

The SMA model, a form that covers the full isotherm shape, is derived from the stoichiometric displacement model (59, 60), which describes just the affinity (linear region). The SMA model accounts for the full isotherm form by allowing for the steric hindrance of bound protein on salt counterion binding (61). Therefore, the model has the advantage of having the salt dependence integrally included. The effect of pH has also been incorporated based on a protein net charge model considering the number of amino acids interacting with the stationary phase (36). The SMA model has been shown to be effective in describing many situations, but deficiencies in predicting elution peaks have been noted at saturated loads, which provide a more stringent test of model effectiveness (62).

Such shortcomings show the sensitivity of column behavior predictions to the details of isotherm shape and have led to efforts to improve the isotherm descriptions. Within the idealized framework of the isotherm models, there is ample room to add rigor, but this is typically done selectively in a fashion that does not necessarily make the enhanced model significantly more valid physically than the original. One avenue to model enhancement is to address thermodynamic ideality by accounting for the protein activity (63). Although this can improve peak prediction for high loads, it includes the introduction of additional parameters to approximate the asymmetric activity coefficient (64). A different example of model elaboration is to allow for the possibility of multistate binding within the SMA framework for the special case of adsorption on tentacular resins (65); this model was able to predict an asymmetric hump in the elution peak.

The uncertainties in postulating model enhancements with a mechanistic basis can be obviated by allowing for empirical modifications in isotherm shape. A systematic empirical interpolation scheme to fit batch isotherm data was successful in capturing the actual elution peak shape (62) and was further extended to pH gradients (66).

A completely different mechanistic basis for modeling protein adsorption in ion exchange has been developed from the colloidal nature of protein molecules. The initial efforts toward this end, in the 1990s, modeled only adsorption affinity using various representations of the protein shape and charge distribution and the charged nature of the adsorbent surface (67–69). However, accounting for protein–protein interactions, primarily steric and electrostatic, allows the full isotherm to be captured (70). Empirical estimation of parameter values using batch isotherm data measurements yields reasonable descriptions of adsorption in single-component (56) and binary (71) systems, but the more stringent requirements of column elution predictions may require additional empirical modifications in the parameter structure, as discussed above for the SMA model.

A similar framework has been employed in a more recent body of work, in which the colloidal calculations were performed within the DLVO (Derjaguin, Landau, Verwey, and Overbeek) theory (72). The protein charge distribution was evaluated from molecular structural information as a

function of pH and salt concentration. This model has been applied to a multicomponent system of monoclonal charge variants (73). A statistical thermodynamic extension that allows for surface coverage effects has been added via calculation of protein–resin and protein–protein interactions (37) and applied to mAb charge variants on two strong cation-exchange resins.

The availability of such a wide variety of models raises the question of whether there is a clear basis for distinguishing among them. The basic isotherm shapes described by these models are qualitatively similar, but the subtle differences in shape, particularly the sharpness or softness, may have very noticeable impacts on elution peak shapes in column simulations. Additional sensitivity of the peak shape results from the salt dependence, which may be integral to the isotherm model, as in the SMA case, or incorporated separately, as in the Langmuir case. This is illustrated in **Figure 2** for three different isotherm models, shown graphically in 3D to include the salt dependence. Viewed from the right, each isotherm at a given concentration has the usual hyperbolic form, but even a comparison of individual isotherms for the Langmuir and SMA cases shows differences in shape. When salt dependence is factored in as well, the precise form of which is somewhat flexible for all but the SMA case, the 3D surfaces differ more appreciably. The 3D surface broadly affects the predicted peak shapes for linear gradient elution (**Figure 3**), generated using the CADET solver for reasonable values of the parameters. These results show the pronounced differences in peak shape that may result, including a split peak in a single-component system, and emphasize the importance of accurately representing the isotherm.

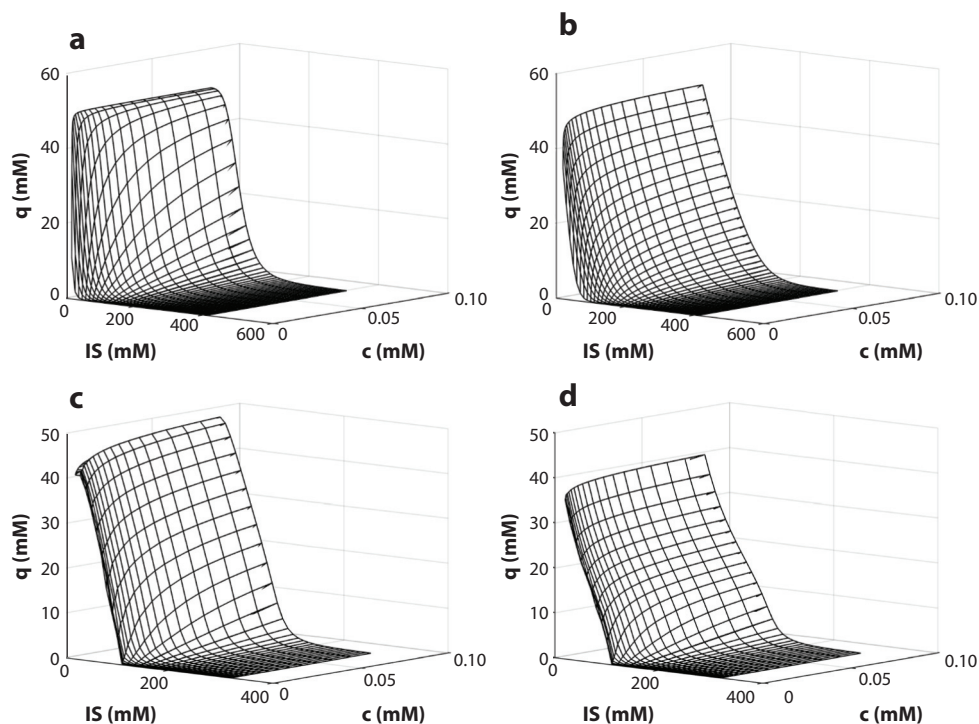


Figure 2

Three-dimensional isotherm shapes for different isotherm models. (a) Langmuir, (b) steric mass action, (c) colloidal, and (d) colloidal with parameters giving rise to split peak elution prediction. Different isotherm surface shapes induce different peak shapes in gradient elution, as shown in **Figure 3**, although the retention times are similar in all cases.

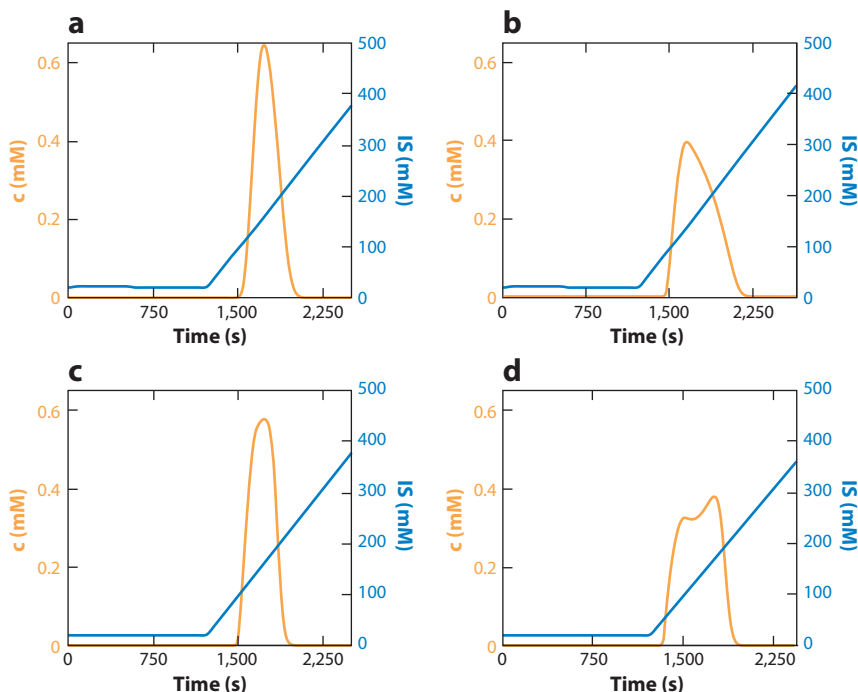


Figure 3

Simulated peak shapes for gradient elution for different isotherms. The peak shapes differ noticeably despite the similar retention times. (a) Langmuir, (b) steric mass action, (c) colloidal, and (d) colloidal with parameters leading to split peak elution prediction. The Langmuir isotherm gives a near-Gaussian peak, whereas the SMA isotherm predicts an asymmetric peak. The colloidal model can produce different peak shapes, including the split peak shown in panel d, even for a single component.

Isotherm Models for Affinity Chromatography

Affinity chromatography is widely used at a laboratory scale, but in large-scale production Protein A (PrA) chromatography is the dominant player, enabling high-selectivity capture of mAbs and mAb-like biotherapeutics (74, 75). The high selectivity of affinity separations and their use for capture rather than polishing has obviated the need for modeling in many cases, but this is changing with the quest to improve the protein loading (76), including in multicolumn configurations that enable high throughput (77). The interaction of PrA with antibodies is a classic biomolecular recognition phenomenon for which the Langmuir binding model is physically appropriate (overlooking the potential for non-1:1 stoichiometry), and indeed most affinity models have used variants of the Langmuir model in view of its ease of implementation (78, 79). For example, Pabst et al. (78) used the Langmuir isotherm without any modification to fit batch isotherm data for 12 different affinity resins and were then able to predict the dynamic binding capacity (DBC) with high accuracy. Elution by the normal route of changing pH was also modeled using a modified Langmuir isotherm, adapted to account for pH effects on the equilibrium binding coefficient (80). Breakthrough curves in PrA have also been modeled successfully using even simpler models, such as a linear isotherm and linear driving force, leading to a $\mathcal{J}$ -function or cumulative exponentially modified Gaussian distribution to approximate breakthrough behavior (81). These outcomes support the notion that affinity chromatography does not generally present a major challenge in protein chromatographic modeling.

Isotherm Models for Hydrophobic Interaction Chromatography

Hydrophobic interaction chromatography (HIC) often has a role in intermediate purification and polishing steps in the downstream processing of biotherapeutics. In mAb processes, this especially involves separation of product-related impurities, such as protein aggregates and fragments, from monomer (82). The surfaces used are mildly hydrophobic (short alkyl chains or phenyl groups), which can be somewhat denaturing but less so than reverse-phase adsorbents, which are consequently used rarely in biomanufacturing; insulin production is a notable exception (83, 84). In HIC, protein binding requires not just the hydrophobic ligands but also mobile-phase modifiers, usually kosmotropic salts (85).

Adsorption isotherms in HIC have the customary hyperbolic form, but they are usually notably softer than those seen in IEX (86), indicating weaker adsorption. As a result, relatively simple isotherm forms such as the Langmuir model are used, and differences more commonly lie in the mechanistic basis sought to explain and model protein-surface affinity—the low-concentration limit of the isotherm. The most widely used retention model in the presence of salt is based on the solvophobic theory (87), in which the two main parameters are the salt molarity and the salt identity, as expressed in the molal surface tension increment. However, the model considers only the protein–ligand interaction and ignores the effect of the salt on the solution properties of the protein; this is inconsistent with the original interpretation of HIC as “salting-out” chromatography (88). Alternative approaches have invoked hydration arguments, applied to the protein, ligand, and/or ions (89, 90), or thermodynamic models such as the preferential interaction model (86, 91) for protein adsorption in HIC. A key factor determining selectivity and elution order is the dependence on the mobile-phase modifier concentration, often a kosmotropic salt, and at low loadings even just a linear isotherm captures much of the essential behavior. Particularly interesting column behavior has been seen and modeled for batch isotherm data showing a U-shaped dependence on the salt concentration, which can lead to readsorption during elution and hence to a peak trapped in the column (92).

Extending the adsorption description from the low-concentration linear region to the full isotherm is straightforward for the soft isotherms characteristic of HIC, as noted above. The standard Langmuir model is often adequate (93, 94), but a quadratic form of the Langmuir isotherm has also been used (86) that was verified in column simulations for both linear-gradient and displacement separations (93). Consideration of the equilibrium between ordered water molecules and bulk-like water molecules around the protein and ligand has also led to a new SMA-like HIC model (90). The effect of the hydrophobic ligands on the protein conformation can affect column behavior, and this has also been modeled, for instance, by combining a protein spreading model with a competitive Langmuir binary isotherm to model monomer and aggregate species of protein (82).

Isotherm Models for Multimodal Chromatography

Multimodal or mixed-mode chromatography (MMC) makes use of more complex ligands, usually with a combination of ion-exchange and hydrophobic interaction character that often give rise to unusual selectivity trends. Because MMC can help reduce the number of unit operations needed in a process, it has gained increased attention for purification of biotherapeutics over the last decade (95), although process development remains challenging and much remains to be done to model MMC reliably. Because of the dual IEX/HIC character of MMC, operation is often under conditions in which one mode is dominant, allowing a relevant IEX or HIC model to be used effectively. For example, under IEX conditions both the SMA isotherm (96, 97) and the Langmuir model (98) have been used successfully. It is, however, possible to combine both modes

into a unified framework; this has been done within Mollerup's thermodynamic framework (99) with parameters included for both salt and pH dependence (100). The model has been further extended to account for the release of water molecules at the apposed hydrophobic interfaces to describe the influence of surface contact area between the protein and the ligand on protein–ligand binding (101). This model also captures the U-shaped dependence on modifier concentration, discussed above for HIC and observed for two mAbs. However, MMC is still a relatively new and incompletely understood mode of chromatography, and the knowledge base of isotherm models is limited.

PARAMETER ESTIMATION

The details of the transport or isotherm model used are often selected based on the ease with which the corresponding parameters can be estimated. The GRM with the inclusion of all the transport terms is the most comprehensive of the models typically used but requires more computing power to simulate the peak shape, an issue that is less significant with current computational capabilities. A more significant issue is possible confounding of effects, leading to correlated parameter values, whether between different transport contributions or between transport and isotherm parameters. This emphasizes the need for independent measurements to estimate at least a subset of the parameters, although limitations of time and materials may be significant constraints during process development. To the extent that column measurements are used to estimate parameter values, it is essential to account for or to minimize experimental effects that are not explicitly included in the column model. For instance, the contribution of the extracolumn volume to dispersion must be taken into account (58, 102, 103). The open-source CADET package allows inclusion of continuous stirred tank reactor and dispersed plug flow models to account for extracolumn dispersion.

Transport Parameters

The GRM contains parameters describing transport processes at the column and the individual bead levels. Some of these can be determined straightforwardly or do not affect the results appreciably, but others are central to the model accuracy. Column axial dispersion is easily estimated from experiments under nonbinding conditions, run at different flow rates; height-of-theoretical-plate methods and the method of moments have been used extensively (104). The external mass-transfer coefficient can be estimated from published correlations, but it is generally not a limiting transport effect at the protein concentrations normally used for preparative chromatography.

Intraparticle effects usually provide the dominant transport resistance and continue to raise questions regarding physical reality. As discussed earlier, the GRM includes separate contributions for pore and surface/homogeneous diffusion. For most GRM implementations in the literature, these two contributions have been lumped into a single effective pore diffusion contribution (58, 65, 105), at least in part to avoid having to estimate two separate diffusivity values. In practice, the pore diffusivity is often estimated from diffusion, porosity, and tortuosity correlations (106–108), but it may also be estimated using the Van Deemter equation under nonbinding conditions. Both of these approaches omit possible contributions to intraparticle transport owing to surface diffusion. Whether surface diffusion actually occurs has been a controversial issue, as pore and surface diffusion often yield similar uptake profiles (109). However, good evidence that surface diffusion does occur emerged with confocal microscopy observations of uptake profiles in individual particles (110, 111). Nevertheless, surface diffusion is often omitted from column models, an omission that can be important for high-load breakthrough curves and for elution simulations, when the protein binding is weaker (112). The omission of surface diffusion is more justifiable during loading under conditions of very strong binding, where the coupling of pore diffusion

with a near-rectangular isotherm gives rise to uptake behavior consistent with the shrinking-core model. The pore diffusivity can then be estimated under uptake conditions from transient concentration profiles inside a resin bead obtained via confocal microscopy (40, 111, 113, 114). Overall, the accurate estimation of D_p and D_s is still a major hurdle in column modeling. A priori estimates of the pore diffusivity, as outlined above, can provide reasonable but not necessarily exact values, but a complete understanding of surface diffusivity and its estimation is needed.

For lumped-parameter models, such as the transport-dispersive and reaction-dispersive models, the situation is nominally simpler because of the smaller number of parameters involved. However, the transport model parameters reflect the convolution of multiple effects and are therefore difficult to estimate a priori, so they are usually estimated, along with isotherm parameters, by fitting of the model to elution data (inverse fitting) (35, 58, 115).

In general, there is some uncertainty in the estimation of transport parameters, but this is not usually a bottleneck in calibrating a column model. Transport within the resin bead is confounded with binding kinetics and adsorption isotherm parameters. For the nonlinear isotherms that must be used under most preparative conditions, the model effects on peak features depend more strongly on the isotherm shape and hence on (possibly varying) mobile-phase conditions. In most of these cases, inverse fits with initial estimates from available correlations or physical models tend to give reasonable values of effective diffusivities.

Isotherm Parameters

Estimation of model parameters for isotherms can be a challenging task, as it is rarely known a priori which isotherm model is most suitable or whether there is an error in the experimental data. The most frequent and direct method is to estimate isotherm parameters from measured batch isotherm data (56, 71, 99). Such measurements are independent of transport behavior if the system is well equilibrated. However, this may take a day or more in some systems, i.e., much longer than any column experiment, which raises the question of whether this true equilibrium isotherm is the appropriate one to use operationally in a column model. If this question is moot, different isotherm models can be fitted to the concentration distribution between the adsorbed and mobile phases to obtain the best model. The method requires many data points and careful measurements, for which a high-throughput robotic liquid handling system is preferable (116). However, this may still be prone to errors owing to the small scale of the measurements.

Accurate measurements can be made with column experiments, but this may require appreciable amounts of material. Alternatively, a smaller number of column experiments may be fitted to model simulations, but this requires deconvolution of transport effects. Yamamoto's linear gradient elution method is widely used to estimate the dependence of the linear region of the isotherm on the modifier concentration for IEX and MMC (38, 101). The method employs a small pulse load with elution at several different gradient slopes, with the parameters determined by plotting the normalized gradient against the modifier concentration at the peak maximum. Because the method is limited to the linear region of the isotherm, the remainder of the isotherm must be determined separately.

Inverse methods estimate the best-fit parameters by minimizing the error between experimental column data and simulated chromatograms (58, 115, 117); these capabilities are integrated into packages such as CADET and ChromX. As noted earlier, such fitting requires modeling of extra-column effects, and different sets of step and gradient elution data are generally needed to find the global minimum (58, 115). The drawback of this method is that, as noted above, parameters such as diffusivities and isotherm parameters are heavily confounded, so constraints should be imposed on parameter values in the fitting algorithms to limit estimates to physically meaningful values.

MODELING APPLICATIONS

Despite the uncertainties outlined above regarding both model formulation and parameter estimation, most physical phenomena in the chromatography of biotherapeutics are largely captured in currently available models. Therefore, depending on the accuracy needed, such modeling can play a role in the design, development, optimization, and control of the chromatography process. The subsections below discuss some recent examples of how the models have been used.

Prediction of Capacity

The DBC is a quantitative measure of adsorbent performance that is widely used by resin manufacturers. The DBC is determined from the protein breakthrough curve and depends on operating factors such as buffer conditions and flow rate. An accurate model of breakthrough can, therefore, be used to optimize the residence time of product loading (78, 79), as well as to assess the value of approaches to increase the DBC, such as the use of dual flow rates (118). Models for breakthrough can also aid selection of adsorbent media based on pressure drop and capacity (78). Models are especially valuable for assessing the potential effect of changes that are more complicated to implement in practice. For example, models for affinity chromatography have been shown to be suitable for designing a multicolumn setup and for simulating continuous operation (81, 119), and a multicomponent IEX model was used to design a multicolumn displacement chromatography scheme for separation of product-related impurities (120). Column models have also been used to determine the extent of resin fouling over its lifetime (121, 122).

Impurity Removal

Whereas manufacturers often use the DBC as a measure of performance during loading, the selectivity or extent of impurity clearance during elution is difficult to generalize for a particular resin applied to different separations. A large body of modeling has therefore been directed to predicting the clearance of key impurities, especially in mAb manufacture, which represents a large fraction of biopharmaceutical manufacturing. Protein aggregates are a major concern and should be limited to ~1–10% in the process stream, and monomer–aggregate mixtures have often served as a prototypical system to validate multicomponent mechanistic column models. Various IEX, HIC, and MMC models have been shown to predict protein monomer and aggregate retention in gradient elution (36, 82, 97, 105, 123). A more challenging separation for mAb processes is that of charge variants, which represent a heterogeneous mix of molecules with different charge distributions; however, these have also been modeled successfully, at least for IEX (37, 73, 115, 117, 120). Much less work has been performed on models of host-cell protein (124) or DNA removal owing to the complexity and often incompletely defined nature of those impurities. For impurities for which accurate models are available, the models can be used to make fine-grained design and operating decisions, such as on product pooling where fractionation is used (105, 115).

Process and Quality Control and Improvement

If the numerical values of model parameters are accurately determined, they can help to predict and understand column behavior. Analysis of parametric sensitivity can aid in verifying estimated parameter values and their impact on simulated chromatograms (125, 126). Statistical methods are also available to assess the quality and uncertainty of the mechanistic models (127). Validated models can then be used to study the impact of sensitive perturbations in parameters such as buffer pH, ionic strength, or gradient during elution. Models can also be used to predict differences between lab- and manufacturing-scale chromatography performance (80). A particularly

thorough model structure in which parameter variations with time are layered on top of the basic model may be used to monitor changing operating conditions to maintain high product yield while optimizing the process performance in terms of capacity utilization, buffer consumption, and resin aging (128).

MODEL LIMITATIONS AND FUTURE PERSPECTIVES

Mathematical models of chromatographic systems, like any other model, provide an imperfect representation of physical reality, but one that is often fully adequate to fulfill the needs of the user. Possible deviations from the idealized model formulation include structural ones, such as particle size heterogeneity in the resin beads; operational ones, such as nonuniform flow profiles; or simply oversimplification of what is ultimately a complex, multiscale system. At one level, such issues can be addressed by building additional features into the model, but more important is the root cause analysis of the reason for the experimentally observed deviation from model-predicted behavior.

An important specific category of such deviations is due to the sometimes unpredictable sensitivity of the structure and properties of biotherapeutic products to conditions encountered during purification. An example mentioned earlier is that of protein conformational changes in HIC, and possibly also in other modes of chromatography. Various other cases have been reported in which unexpected system behavior is encountered, such as two elution peaks for a single-component system (129). Modern tools that enable surface mapping, such as hydrogen–deuterium exchange mass spectrometry, are used to study possible protein instability during the chromatography process (130, 131). Once better understood, such complicated systems can be modeled in greater detail (132, 133).

In the discussion earlier of isotherm models, the interplay was noted between the usual continuum representation of the full isotherm and the molecular-level reasoning sometimes employed to describe protein–surface or protein–ligand affinity. This type of reasoning often leads to a conceptual model that can capture one or more important features of experimental isotherms, but it does not usually provide a complete molecular-level picture of interaction at the binding site. Molecular modeling can provide insights into protein–ligand complexes down to the atomistic level, making available information that can be difficult to obtain from experimental studies. Models have been presented for traditional affinity interactions (134), IEX resins (135), and MMC (136–138). Numerous significant obstacles remain, including quantitative validation of simulation results and translation of simulation data into an adsorption representation that can be integrated into an effective column model. However, the molecular-level understanding that is gleaned using such tools can help in formulating more robust and mechanistic isotherm models.

As discussed earlier, there remain uncertainties in representing both the model form and the parameter values for protein diffusion in porous beads, especially for larger molecule-to-pore size ratios. Tools like confocal microscopy have been employed to study protein uptake kinetics at the bead level and to infer therefrom the mechanisms involved. Such insights can be used along with chromatography models to model column behavior (98, 139). Transport properties may also be obtained by molecular simulations on different scales, for cases in which stationary-phase structural details are available (140).

In stark contrast to detailed modeling efforts at the molecular, continuum, or column levels are black-box models and machine-learning approaches such as artificial neural networks; these may be used independently or in tandem (141–143). Artificial neural networks can be a useful tool in both process optimization and parameter estimation when alternative methods fail. The availability of large sets of experimental process data for many processes that have been operated

successfully at the manufacturing scale for many years may allow such empirical relationships to be used prospectively in lieu of mechanistic models. At the same time, the extensive attention being paid currently to making mechanistic models more accurate and predictive are likely to yield improvements in process development, characterization, and control over the next decade.

CONCLUSIONS

The chromatographic models that are in current use differ little in overall structure from those that evolved during the mid-twentieth century. The real impetus for the intensive attention such modeling is now receiving for biopharmaceutical applications is due to the increased availability of computational partial differential equation solvers specifically packaged for such problems, as well as the industrial demand for model-based support. Although some aspects of modeling both isotherms and transport remain incompletely understood, the growing body of research in this area is informing both model construction and model interpretation. As this body of research becomes augmented by molecular understanding and rapidly growing databases of empirical data, there is considerable promise that process development, automation, and control of biopharmaceutical chromatographic processes can attain the same degree of underlying quantitative understanding as is now routine for many unit operations in the traditional chemical process industries.

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Water Treatment: Are Membranes the Panacea?

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Errata

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